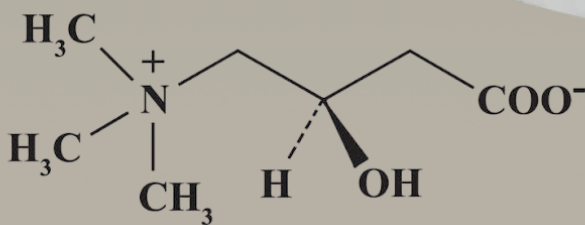


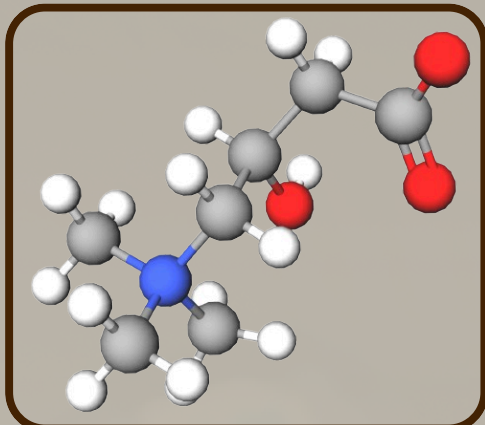
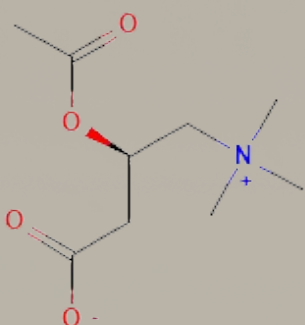


CHEMICAL IDENTITY

- **IUPAC Name:** (3R)-3-hydroxy-4-(trimethylazaniumyl)butanoate
- **Molecular Formula:** $C_7H_{15}NO_3$
- **Molar Mass:** 161.20 g/mol
- **CAS No.:** 541-15-1
- **PubChem CID:** 10917



STRUCTURE (L-Carnitine)



INTERESTING FACT

L-carnitine plays a crucial role in transporting long-chain fatty acids into mitochondria, where they can undergo β -oxidation to produce energy.

L-carnitine temporarily forms acylcarnitines with fatty-acyl groups. This allows long-chain fatty acids to cross the inner mitochondrial membrane before being released for β -oxidation.

KEY PROPERTIES & DATA

Appearance: White crystalline solid

Molecular weight: 161.20 g/mol

Formula: $C_7H_{15}NO_3$

Hydrogen-bond donors: 1

Hydrogen-bond acceptors: 3

Topological polar surface area: 60.4 Å²

Chemical character: Water-soluble, zwitterionic compound

USES & VALUE

Biological role

- Plays an essential role in cellular energy metabolism.
- Helps support energy production from fatty acids.
- Used medically to manage certain forms of carnitine deficiency.

Commercial/consumer use

- Commercially produced for use in supplement and pharmaceutical applications.
- Its value comes from its established role in fatty-acid metabolism and its applications in specific nutritional and medical products.



SOURCES

Red meat, Dairy Products (Milk, Butter, Cream), Supplements and Skin Care Products



RISKS AND SAFETY

- Carnitine naturally occurs in foods and the human body.
- Excessive supplemental intake can cause nausea, vomiting, diarrhea, and a fishy body odor.
- Certain medical conditions may increase the risk of adverse effects.
- Also investigated the conversion of carnitine by gut bacteria into trimethylamine N-oxide (TMAO); health implications remain an area of research.

CHEMICAL ENGINEERING CONNECTION

- **Reaction Engineering:** Optimizes conditions for L-carnitine production.
- **Separation and Purification:** Removes impurities and unwanted reaction products.
- **Crystallization:** Can be used to recover and purify the solid product.
- **Filtration & Drying:** Separates crystals and removes residual moisture.
- **Mass Balances:** Tracks reactants, products, and waste throughout the process.
- **Process Optimization:** Improves yield, purity, and production efficiency.
- **Quality Control:** Ensures chemical identity, purity, and product consistency.

